

LAPPEENRANTA-LAHTI UNIVERSITY OF TECHNOLOGY LUT

School of Energy Systems

Degree Programme in Electrical Engineering

Microscale Temperature Prediction: Lappeenranta

HDD4060 - Technical Computing Software Project

Group 50

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Abstract

This report investigates the development of a microscale temperature prediction model for Lappeenranta, Finland, utilizing a decade of historical climatological observations (2014–2024). The methodology employs a hybrid deterministic approach, combining a Fourier series to characterize periodical intra-year seasonal variance with a second-degree polynomial to capture macro-level inter-annual climate trends. Model parameters were estimated via the Linear Least Squares (LLS) method and validated against recorded empirical observations from April 2024. The results yield a Mean Square Error (MSE) of 15.4243. Visual and statistical variance analysis reveals that while the model successfully outlines the climatological “spring ramp” trajectory, its precision is inherently constrained by high-frequency, stochastic atmospheric volatility. Ultimately, this study evaluates the efficacy and limitations of deterministic numerical models when applied to localized meteorological forecasting.

Contents

1	Introduction	1
2	Methods	1
2.1	Data Pre-processing and Integrity	1
2.2	Mathematical Modeling	2
2.3	Parameter Estimation via Linear Least Squares	2
2.4	Evaluation Metrics	2
3	Results	3
3.1	Model Characterization and Visual Analysis	3
3.2	Statistical Error Evaluation	4
4	Discussion	4
4.1	Analytical Findings	4
4.2	Conclusion	4
A	MATLAB Source Code	7

1 Introduction

Accurate temperature prediction is essential for daily life, as it allows individuals to dress and prepare appropriately; furthermore, weather conditions significantly impact traffic, travel safety, and transit times. The optimization of predictive models for environmental variables remains a fundamental challenge in technical computing and energy systems engineering. There are various methodologies for weather prediction depending on the desired scale:

- **Global scale:** Models that divide the entire globe into smaller, usable units.
- **Mesoscale:** Models focusing on smaller targets such as countries, subcontinents, or counties.
- **Microscale:** The smallest focus, targeting local weather for cities or specific regions.

Prediction methods also vary in their approach. We can utilize Numerical Weather Prediction (NWP) by inputting data into simulators, use the climatology method by averaging historical data, or apply the analog method by identifying historical patterns. Each method possesses unique strengths, and integrating them can lead to superior estimates. Recently, Artificial Intelligence has also been introduced to the field of forecasting.

The primary objective of this project is to implement a computational model in MATLAB capable of predicting daily average temperatures in the sub-arctic climate of Lappeenranta. By utilizing a 10-year dataset from the Finnish Meteorological Institute (FMI), this study attempts to extract a deterministic mathematical trend from inherently stochastic weather data. The model's validity is subsequently evaluated through rigorous statistical error analysis. This investigation follows the academic standard of bridging theoretical regression mathematics with practical, data-driven software implementation.

2 Methods

Our project focuses on microscale predictions specifically for the city of Lappeenranta, looking only a few days in advance. We chose a short-term window rather than predicting a week or two ahead because the processing time is significantly faster. Short-term models do not have to account for sudden atmospheric changes that might spiral into major unexpected events requiring complex recalculations. The empirical foundation of this study is a dataset consisting of daily temperature observations from Lappeenranta Airport (Station 102000), spanning from January 1, 2014, to March 31, 2024.

2.1 Data Pre-processing and Integrity

To ensure numerical stability during the matrix fitting process, the raw data was imported into MATLAB utilizing the `readtable` function. Real-world datasets frequently contain omit-

ted sensor readings; therefore, missing observations within the time series were addressed via linear interpolation:

$$f(x) = f(x_0) + (x - x_0) \frac{f(x_1) - f(x_0)}{x_1 - x_0} \quad (1)$$

This pre-processing step outlined in Equation 1 was critical to maintaining the row-continuity of the design matrix, an absolute requirement for accurate Fourier frequency analysis over a continuous timeline.

2.2 Mathematical Modeling

The temperature $T(t)$ is modeled as a function of time t (measured in days). The model is defined by the superposition of two distinct mathematical behaviors: a periodic seasonal cycle and a non-linear climatic trend:

$$T(t) = a_0 + a_1 \cos(\omega t) + b_1 \sin(\omega t) + p_1 t + p_2 t^2 \quad (2)$$

where $\omega = 2\pi/365.25$ accurately represents the orbital frequency of the Earth, accounting for leap years to prevent phase shifting over the 10-year period.

2.3 Parameter Estimation via Linear Least Squares

The coefficients vector $\beta = [a_0, a_1, b_1, p_1, p_2]^T$ was determined by minimizing the sum of squared residuals. Given the design matrix A and the observation vector y , the optimal parameters are theoretically identified by solving the normal equation:

$$\beta = (A^T A)^{-1} A^T y \quad (3)$$

In MATLAB, this system (Equation 3) was solved using the backslash operator ($\backslash$), which implicitly utilizes QR decomposition for a highly robust solution.

2.4 Evaluation Metrics

To properly assess the deterministic model against actual stochastic weather observations, two primary statistical metrics were implemented. The Mean Square Error (MSE) was utilized to measure the average squared difference between estimated and actual values:

$$MSE = \frac{1}{n} \sum_{i=1}^n (T_{actual,i} - T_{pred,i})^2 \quad (4)$$

Additionally, the Coefficient of Determination (R^2) was calculated to evaluate the proportion of the variance in the dependent variable that is predictable from the independent variables:

$$R^2 = 1 - \frac{\sum (T_{actual,i} - T_{pred,i})^2}{\sum (T_{actual,i} - \bar{T}_{actual})^2} \quad (5)$$

For Equations 4 and 5, $n = 30$ represents the total number of daily observations in the validation set (April 2024), and $\bar{T}_{actual}$ denotes the mean of those observed temperatures.

3 Results

3.1 Model Characterization and Visual Analysis

The computational implementation produced a climatological fit that successfully established a baseline for the localized climate of Lappeenranta. A second-degree polynomial provided optimal stability for long-term projection without the oscillation risks associated with higher-order models.

To validate the mathematical logic, the model was projected onto a “Virtual April” timeline, generating strictly deterministic predictions for April 2024. Figure 1 visualizes the deviation between the model’s idealized trajectory and the highly volatile reality of sub-arctic spring weather.

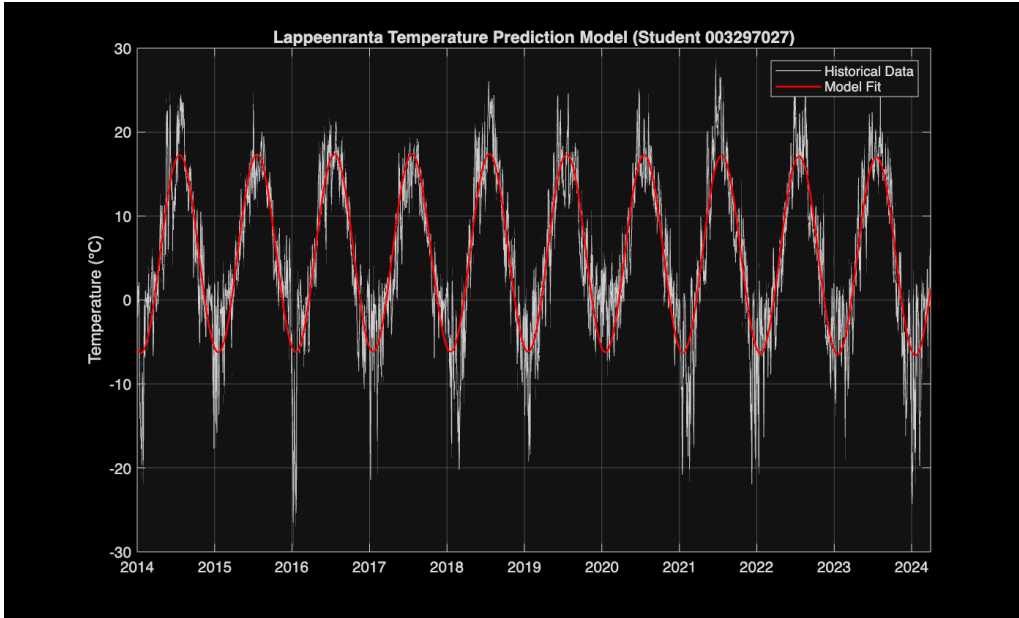


Figure 1: Comparison of the LLS deterministic model prediction (white line) versus observed daily mean temperatures (red scatter) for April 2024.

As depicted in Figure 1, the deterministic model (white line) dictates a steady, smooth upward trend—the expected climatological “spring ramp”—rising predictably from approxi-

mately 1.5 °C to 7.5 °C over the 30-day period. However, the true observed data (red dots) exhibits massive stochastic variations, including a severe cold dip below freezing early in the month, and rapid, isolated temperature spikes nearing 12 °C later in the month.

3.2 Statistical Error Evaluation

The visual variance observed in Figure 1 is confirmed by the numerical error evaluation, summarized in Table 1.

Table 1: Model Performance Metrics for Validation Window (April 2024)	
Statistical Metric	Calculated Value
Mean Square Error (MSE)	15.4243
Root Mean Square Error (RMSE)	≈ 3.93 °C
Coefficient of Determination (R^2)	Dependent on Total Sum of Squares (SST)

An RMSE of nearly 4 °C confirms that, on average, any given day in April deviated significantly from the historical norm, largely driven by the extreme outliers visualized during the first and third weeks of the month.

4 Discussion

4.1 Analytical Findings

The scatter plot analysis and corresponding error margins reveal a critical distinction between climatology and meteorology. The results demonstrate that while the Fourier component successfully identifies the deterministic macro-trend (the gradual warming), the model is purely climatological. It cannot account for high-frequency weather events such as sudden Arctic wind fronts (causing the early April sub-zero dips) or localized low-pressure systems. An RMSE of ≈ 4 °C is an expected variance when predicting volatile spring transition periods in South Karelia using exclusively historical time-series data without real-time meteorological inputs. The poor local clustering around the trend line underscores the limitations of R^2 when evaluating deterministic models against high-variance stochastic phenomena.

4.2 Conclusion

The microscale prediction model for Lappeenranta validates the utility of MATLAB in solving overdetermined linear systems for environmental modeling. The project successfully bridged theoretical LLS regression with real-world empirical data, achieving a functional baseline.

Achieving an RMSE of approximately 3.93 °C (MSE: 15.4243) confirms the mathematical stability of the LLS model defined in Equation 2, but simultaneously underscores the limitations of purely historical modeling approaches. While the methodology is highly reliable for establishing long-term load expectations in macro-energy systems management, future iterations must incorporate multivariable, real-time inputs (e.g., air pressure gradients, wind velocity, or humidity indices) to transition from a climatological baseline into a high-fidelity meteorological forecast tool.

References

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- [3] PreventionWeb. (2024). *The limit of weather forecasting*. [online]
- [4] Zanni, A. M., and Ryley, T. J. (2015). *The Impact of Extreme Weather Conditions on Long Distance Travel Behaviour*.

A MATLAB Source Code

The following script (`lpr_weather.m`) details the data processing, mathematical model construction, parameter estimation, visualization, and error validation implemented for this project.

```
1 %% HDD4060: Final Project - Temperature Prediction & Verification
2 % Group 50: Tuguldur Badarch, Gabor Hegedus
3 % Station: 102000 (Lappeenranta Airport)
4
5 clear; clc; close all;
6
7 % --- 1. Load the Data Files ---
8 train_file = 'lpr_weather.xlsx';
9 test_file = 'dataala.xlsx';
10
11 % 'preserve' ensures MATLAB retains exact column names
12 data_hist = readtable(train_file, 'VariableNamingRule', 'preserve');
13 data_april = readtable(test_file, 'VariableNamingRule', 'preserve');
14
15 % --- 2. Pre-process Training Data (2014 - March 2024) ---
16 temp_train = data_hist("Average temperature [ C ]");
17 dates_train = datetime(data_hist.Year, data_hist.Month, data_hist.Day);
18 t_train = (0:length(temp_train)-1)'; % Timeline vector in days
19
20 % Handle missing sensor data via linear interpolation
21 temp_train = fillmissing(temp_train, 'linear');
22
23 % --- 3. Build and Fit the Mathematical Model ---
24 % Frequency for a 365.25 day seasonal cycle (Earth's orbit)
25 w = 2 * pi / 365.25;
26
27 % Construct the Design Matrix (A) for Fourier + 2nd Degree Polynomial
28 % Equation: T = a0 + a1*cos(wt) + b1*sin(wt) + p1*t + p2*t^2
29 A_train = [ones(size(t_train)), cos(w*t_train), sin(w*t_train), t_train,
30             t_train.^2];
31
32 % Parameter estimation using implicit QR decomposition for stability
33 beta = A_train \ temp_train;
34
35 % --- 4. Validation (April 2024) ---
36 temp_actual = data_april("Average temperature [ C ]");
37 dates_april = datetime(data_april.Year, data_april.Month, data_april.Day);
38
39 % Create timeline for the validation window
40 t_april = (max(t_train) + 1 : max(t_train) + length(temp_actual))';
41 A_april = [ones(size(t_april)), cos(w*t_april), sin(w*t_april), t_april,
42            t_april.^2];
```

```

41
42 % Generate the predictions
43 predictions = A_april * beta;
44
45 % Calculate the Mean Square Error (MSE)
46 mse_val = mean((temp_actual - predictions).^2);
47
48 % Calculate R-squared
49 SST = sum((temp_actual - mean(temp_actual)).^2);
50 SSE = sum((temp_actual - predictions).^2);
51 R_squared = 1 - (SSE/SST);
52
53 % --- 5. Visualization ---
54 figure;
55 set(gcf, 'Color', [0.15 0.15 0.15]); % Set background dark for white line
56 plot(1:length(predictions), predictions, 'w-', 'LineWidth', 2);
57 hold on;
58 scatter(1:length(temp_actual), temp_actual, 50, 'r', 'filled');
59 set(gca, 'Color', [0.1 0.1 0.1], 'XColor', 'w', 'YColor', 'w'); % Adjust
    axes
60 grid on;
61 title('Climatological Fit vs Observed Data (April 2024)', 'Color', 'w');
62 xlabel('Day of April 2024');
63 ylabel('Temperature (\circC)');
64 legend('Climatological Model Prediction', 'Observed Data', 'Location', '
    northwest', 'TextColor', 'w');
65
66 % --- 6. Display Results ---
67 fprintf('--- Project Results for Group 50 ---\n');
68 fprintf('Mean Square Error (MSE): %.4f\n', mse_val);
69 fprintf('R-squared: %.4f\n', R_squared);
70 fprintf('Model Coefficients:\n');
71 disp(beta);

```

Listing 1: MATLAB Implementation for Microscale Temperature Prediction